

*Power Transformer Differential Relay
Protection Using MATLAB/Simulink*

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Abstract

This project presents a MATLAB/Simulink model for a power transformer differential protection relay. The main purpose of the project is to protect a three-phase power transformer against internal faults by comparing the currents on both sides of the transformer. The simulated transformer is rated at 200 MVA, 11 kV / 33 kV, 50 Hz, with grounded-wye connections on both sides.

The relay algorithm is designed using a MATLAB Function block. It calculates the differential current and the restraint current then decides whether the fault is inside or outside the protected zone. The relay also includes practical features such as percentage differential protection, a dual-slope characteristic, second harmonic blocking for inrush current, fifth harmonic restraint for overexcitation high set differential tripping, trip latching and circuit breaker control.

The system was tested under five scenarios: normal operation, three-phase-to-ground internal fault, external fault, transformer energization with inrush current, and single-phase-to-ground internal fault. The simulation results showed that the relay did not trip during normal operation, external faults, or inrush current conditions. It successfully operated during internal faults and sent an opening command to the circuit breaker (CB).

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1. Introduction

Power transformers are important components in electrical power systems because they transfer power between different voltage levels. They are also expensive and critical for system operation, so protecting them from faults is very important. A transformer fault can cause serious equipment damage and may affect the stability and continuity of the power system [1].

One of the most common protection methods used for power transformers is differential protection. The main idea is to compare the current entering the protected zone with the current leaving it. Under normal operating conditions, these currents should be almost balanced after considering the transformer ratio and current direction. In this case the differential current remains small and the relay should not trip [1, 2].

When an internal fault occurs inside the protected zone the current balance between the two sides of the transformer is disturbed. As a result, the differential current increases. If this current becomes higher than the relay setting, the relay sends a trip signal to open the circuit breaker (CB) and isolate the faulted transformer. This makes differential protection a fast and selective method for transformer internal fault detection [2, 3].

Transformer differential protection is not always straightforward. Some conditions may produce high currents even when there is no internal fault. A common example is magnetizing inrush current which appears when the transformer is energized. This current can be large and may cause false tripping if the relay does not recognize it correctly. For this reason, second harmonic blocking or restraint is commonly used since transformer energization can produce a large inrush current with a noticeable second harmonic component [3, 4].

Another condition that should be considered is transformer overexcitation. Harmonic restraint and blocking methods are commonly used in transformer differential relays to improve relay security and reduce the possibility of unwanted operation during non-fault conditions [4, 5].

2. System Description

2.1 Complete System Model

Figure 1 below shows the complete MATLAB/Simulink model of the proposed transformer differential protection system. The model consists of a three-phase source, CB, a three-phase two-winding transformer, current measurement blocks, fault blocks, and a three-phase load. The transformer is rated at 200 MVA, 11 kV / 33 kV, 50 Hz with grounded-wye connections on both sides.

Current measurements from the primary and secondary sides are sent to a relay subsystem implemented using a MATLAB Function block. The relay calculates the differential current, restraint current, and harmonic content to detect internal faults and distinguish them from external faults and transformer inrush conditions. The protection algorithm also includes second harmonic blocking to improve relay security during transformer energization and prevent unwanted tripping caused by magnetizing inrush current.

In addition, several scopes and display blocks are connected throughout the model to monitor the measured currents, relay outputs, harmonic behavior and CB status during the simulation process. The relay generates a trip signal and a CB command when an internal fault is detected, allowing the transformer to be isolated from the power system quickly.

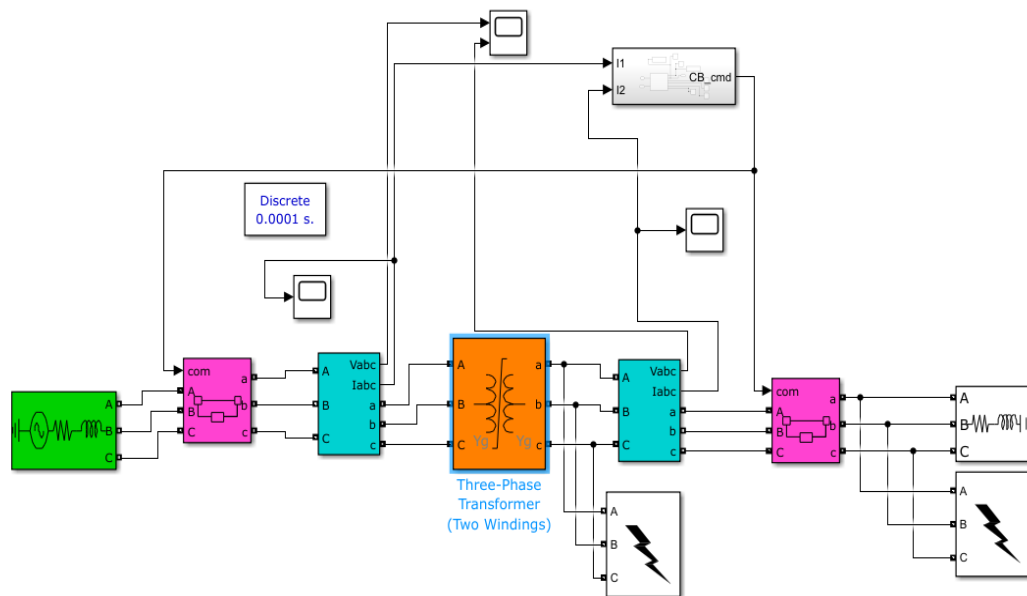


Figure 1. Power transformer differential protection system.

2.2 Relay Subsystem

Figure 2. shows the internal structure of the differential relay subsystem. The relay receives two three-phase current signals, I_1 and I_2 from the current measurement blocks installed on both sides of the transformer. These signals are used as the main inputs to the MATLAB Function block where the protection algorithm is implemented.

Inside the relay algorithm the measured currents are processed to calculate the differential current (I_d), restrain current (I_r) and second harmonic ratio (H_2). The differential current is used to detect internal faults while the restrain current improves relay stability during high-current conditions and external faults. The second harmonic ratio is used to identify transformer inrush current and prevent false tripping during transformer energization.

The relay produces several output signals for monitoring and control. The Trip signal indicates whether the relay has detected an internal fault. The Circuit Breaker command (CB_{cmd}) signal is sent to the CB where a value of 1 keeps the breaker closed and a value of 0 opens the breaker. Other outputs such as $I_d_display$, $I_r_display$, $H2_display$, Inrush Block and inrush current are connected to scopes and displays to monitor the relay behavior during different test cases.

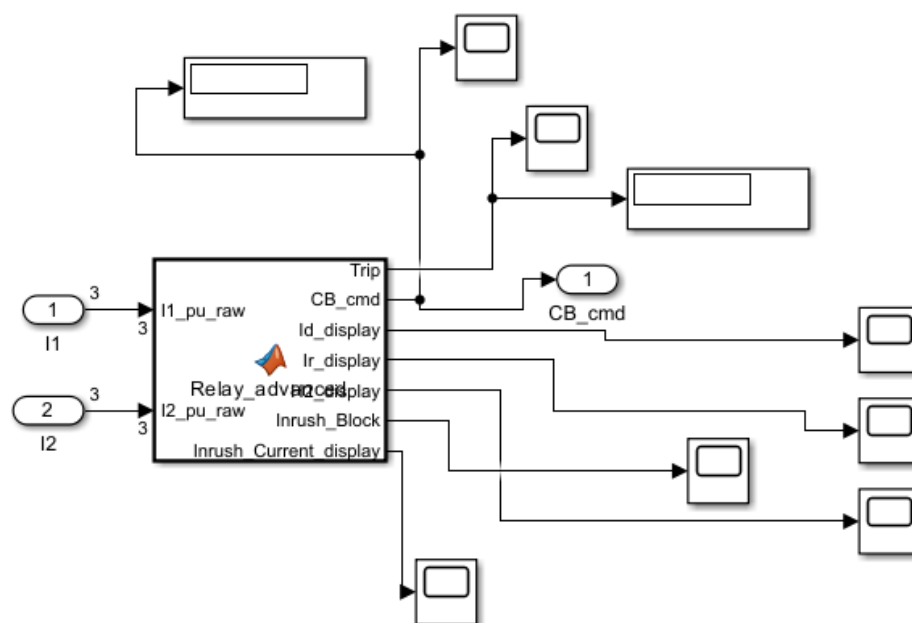


Figure 2. Differential relay subsystem.

3. Methodology

3.1 Discrete Fourier Transform Calculation

In this project, a one-cycle Discrete Fourier Transform is used to extract the fundamental, second harmonic, and fifth harmonic components of the measured current signals. DFT-based harmonic extraction is commonly used in digital relay algorithms to analyze the frequency components of transformer currents [1, 4].

$$I_h = \frac{\sqrt{2}}{N} \sum_{n=0}^{N-1} i(n) e^{-j2\pi hn/N} \quad (1)$$

where I_h is the RMS phasor of harmonic order h , N is the number of samples per cycle and $i(n)$ is the sampled current signal. In this project, $h = 1$ is used for the fundamental component, $h = 2$ for the second harmonic component and $h = 5$ for the fifth harmonic component.

3.2 Secondary Current Compensation

Before calculating the differential current, the secondary-side current is compensated using a polarity correction factor. This ensures that the primary and secondary currents are compared with the correct direction during the healthy operating period.

$$I_{2c,ph} = pol \times I_{2,ph} \quad (2)$$

where $I_{2c,ph}$ is the compensated secondary-side current of each phase, $I_{2,ph}$ is the measured secondary-side current, and pol is the polarity correction factor used by the relay algorithm.

3.3 Differential Current Calculation

The differential current is the main operating quantity used in transformer differential protection. It represents the difference between the currents entering and leaving the transformer protection zone. Under normal operating conditions, the primary and secondary currents are approximately equal after compensation resulting in a very small differential current. During an internal fault, this balance is disturbed and the differential current increases, causing the relay to operate [1, 2].

$$I_{d1,ph} = | I_{1,1,ph} - I_{2c,1,ph} | \quad (3)$$

where $I_{d1,ph}$ is the fundamental differential current of each phase, $I_{1,1,ph}$ is the fundamental primary-side current phasor and $I_{2c,1,ph}$ is the compensated fundamental secondary-side current phasor.

3.4 Restraint Current Calculation

The restraint current is used to improve the stability and security of the differential relay during external faults and high through-current conditions. It helps prevent unwanted relay operation caused by current transformer mismatch, measurement errors, or transient conditions [1, 2].

$$I_{r,ph} = \frac{| I_{1,1,ph} | + | I_{2c,1,ph} |}{2} \quad (4)$$

where $I_{r,ph}$ is the restraint current of each phase.

3.5 Second Harmonic Blocking

The second harmonic component is used to detect transformer magnetizing inrush current. During transformer energization, the inrush current may produce a high differential current even though no internal fault exists. Therefore, second harmonic blocking is used to prevent false tripping during transformer energization [3, 4].

$$I_{d2,ph} = | I_{1,2,ph} - I_{2c,2,ph} | \quad (5)$$

$$H_{2,ph} = \frac{I_{d2,ph}}{I_{d1,ph}} \quad (6)$$

where $I_{d2,ph}$ is the second harmonic differential current, and $I_{d1,ph}$ is the fundamental differential current. If the second harmonic ratio exceeds the selected threshold, the relay identifies the condition as transformer inrush current and blocks the normal differential trip.

In this project, the second harmonic blocking condition is:

$$H_{2,ph} > 0.15$$

3.6 Fifth Harmonic Blocking

The fifth harmonic component is used to improve relay security during transformer overexcitation conditions. Harmonic blocking techniques are commonly used in transformer differential relays to distinguish abnormal operating conditions from internal faults [4, 5].

$$I_{d5,ph} = | I_{1,5,ph} - I_{2c,5,ph} | \quad (7)$$

$$H_{5,ph} = \frac{I_{d5,ph}}{I_{d1,ph}} \quad (8)$$

where $I_{d5,ph}$ is the fifth harmonic differential current for each phase, and $I_{1,5,ph}$ and $I_{2c,5,ph}$ are the fifth harmonic components of the primary and compensated secondary currents. If $H_{5,ph}$ exceeds 0.25, the relay blocks the normal differential trip to avoid false operation during overexcitation.

The fifth harmonic blocking condition is:

$$H_{5,ph} > 0.25$$

3.7 Dual-Slope Percentage Differential Characteristic

The relay uses a dual-slope percentage differential characteristic. This characteristic improves relay sensitivity for internal faults and increases relay stability during high-current external faults. Percentage differential protection is widely used in transformer differential relays to maintain both sensitivity and security [1, 2].

When the restraint current is less than or equal to the break point:

$$I_{r,ph} \leq I_{r,break} \quad (10)$$

the operating threshold is:

$$I_{d,threshold} = I_{pickup} + Slope_1 I_{r,ph} \quad (11)$$

When the restraint current is greater than the break point:

$$I_{r,ph} > I_{r,break}$$

the operating threshold is:

$$I_{d,threshold} = I_{pickup} + Slope_1 I_{r,break} + Slope_2 (I_{r,ph} - I_{r,break}) \quad (12)$$

The relay operates when:

$$I_{d1,ph} > I_{d,threshold} \quad (13)$$

The relay settings used in this project are:

$$I_{pickup} = 0.25 \text{ pu}$$

$$Slope_1 = 0.30$$

$$Slope_2 = 0.70$$

$$I_{r,break} = 1.50 \text{ pu}$$

3.8 High-Set Differential Trip

A high-set differential element is included to provide fast tripping for severe internal faults. This element allows the relay to operate quickly when the differential current becomes very large.

$$I_{d1,ph} > I_{HighSet} \quad (14)$$

In this project:

$$I_{HighSet} = 6.00 \text{ pu}$$

If the differential current exceeds this value, the relay issues a trip command.

3.9 Final Trip Logic

The final relay trip decision depends on the differential operating condition, harmonic blocking conditions, high-set condition, and confirmation time.

$$Trip = 1 \text{ if } (I_{d1,ph} > I_{d,threshold}) \text{ and } (H_{2,ph} < 0.15) \text{ and } (H_{5,ph} < 0.25) \quad (15)$$

The relay can also trip if the high-set condition is satisfied:

$$I_{d1,ph} > 6.00 \text{ pu}$$

To avoid unwanted operation due to short transient disturbances, the operating condition must remain active for a confirmation time of 5 ms before the trip signal is latched.

3.10 Circuit Breaker Command

The circuit breaker command is generated from the trip signal using the following equation:

$$CB_{cmd} = 1 - Trip \quad (16)$$

When $Trip = 0$, then $CB_{cmd} = 1$, which means that the circuit breaker remains closed. When $Trip = 1$, then $CB_{cmd} = 0$, which means that the circuit breaker opens and isolates the transformer from the system.

3.11 Relay Settings table

Parameter	Value
Sampling time T_s	$1 \times 10^{-4} \text{ s}$
System frequency	50 Hz
Samples per cycle	200
I_{pickup}	0.25 pu
Slope1	0.30
Slope2	0.70
I_{rbreak}	1.50 pu
$I_{Highset}$	6.00 pu
H_2	0.15
H_5	0.25
Confirmation time	5 ms
Startup blocking time	0.08 s

4. Differential Relay Operating Principle

The transformer differential relay operates by comparing the currents entering and leaving the protected transformer zone. In this project these currents are measured from the primary and secondary sides of the transformer using current measurement blocks in Simulink. Under normal operating state and external faults, the two measured currents remain approximately balanced after ratio compensation. The differential current remains very small and the relay does not operate [1, 2].

During an internal fault inside the transformer protection zone, the balance between side currents is disturbed. As a result, the calculated side and secondary the primary differential current increases above the relay operating threshold. In the developed Simulink model this condition causes the relay to generate a trip signal and changes the CB command from 1 to 0 which opens the breaker and isolates the transformer from the power system [1, 3].

To improve relay security during transformer energization second harmonic blocking is included in the protection algorithm. This feature is important because magnetizing inrush current may produce a high differential current even when there is no internal fault. Since inrush current usually contains a noticeable second harmonic component, the relay uses the second harmonic ratio to block unnecessary tripping during transformer energization [3, 4].

5. Relay Algorithm

The relay algorithm was implemented using a MATLAB Function block in Simulink. It receives three-phase current signals from both sides of the transformer and processes them to detect internal faults. The algorithm calculates the differential current restraint current and harmonic components then uses these values to decide whether the relay should trip or remain stable.

5.1 Relay Operating Steps

The operating sequence of the relay algorithm is as follows:

1. Measure the currents

The relay receives I_1 and I_2 , which represent the primary-side and secondary-side three-phase currents in per-unit.

2. Store the samples

The current samples are stored in a one-cycle buffer to allow proper signal processing.

3. Extract current components

A Discrete Fourier Transform (DFT) is used to extract the fundamental, second harmonic, and fifth harmonic components.

4. Calculate protection quantities

The relay calculates the differential current I_d , restraint current I_r and second harmonic ratio H_2 .

5. Check the inrush condition

If the second harmonic ratio exceeds the selected threshold, the relay activates the inrush blocking signal.

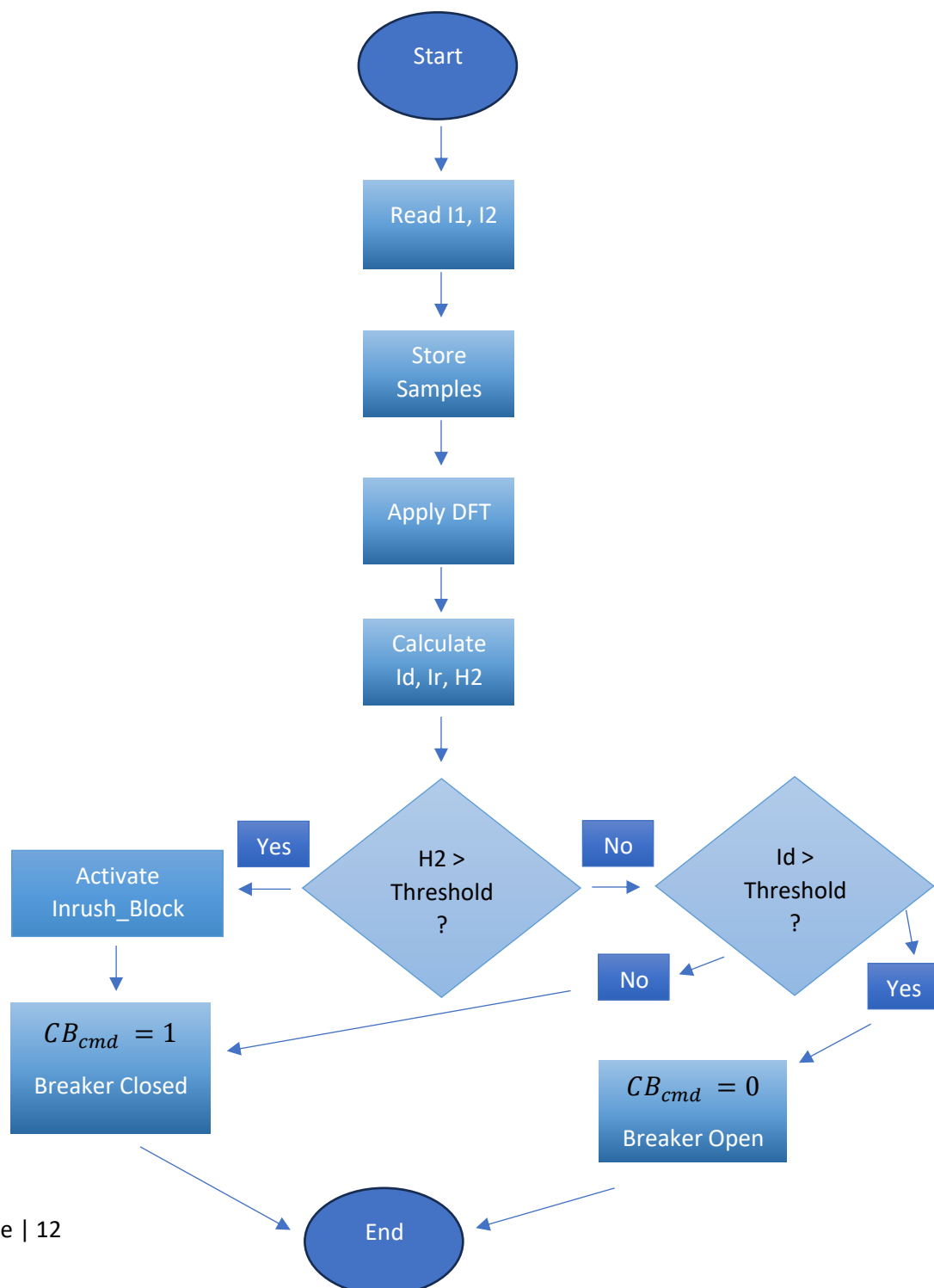
6. Make the trip decision

If I_d is greater than the operating threshold and inrush blocking is inactive, the relay generates a trip signal.

7. Send the breaker command

When the relay trips the CB_{cmd} changes from 1 to 0 causing the CB to open and isolate the transformer.

5.2 Relay Algorithm Flowchart



6. Scenarios and Results

The relay was tested under five conditions: normal operation, three-phase-to-ground internal fault, external fault, transformer energization with inrush current, and single-phase-to-ground internal fault. For the three-phase internal and external fault tests, a three-phase-to-ground fault was applied by selecting phases A, B, C, and Ground in the Three-Phase Fault block. For the single-phase-to-ground internal fault test, only Phase A and Ground were selected. The main monitored signals were I_1 , I_2 , I_d , Trip, Circuit Breaker command (CB_{cmd}), H_2 , and Inrush Block.

6.1 Normal Operation

During normal operation no fault was applied. The primary and secondary currents were almost balanced, so the differential current is very small. Trip stayed 0 and CB_{cmd} stayed 1, which means the circuit breaker remained closed.

Figure 3. shows the primary-side current waveform during normal operation. The current remains stable and balanced because no fault is applied.

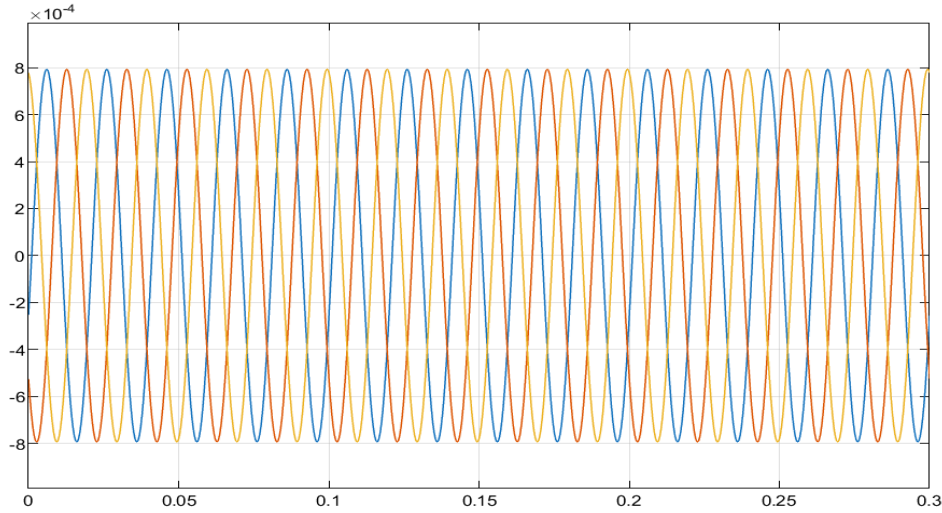


Figure 3. Primary-Side Current during Normal Operation.

Figure 4. presents the secondary-side current waveform during normal operation. The waveform is almost balanced with the primary-side current, which indicates normal transformer operation.

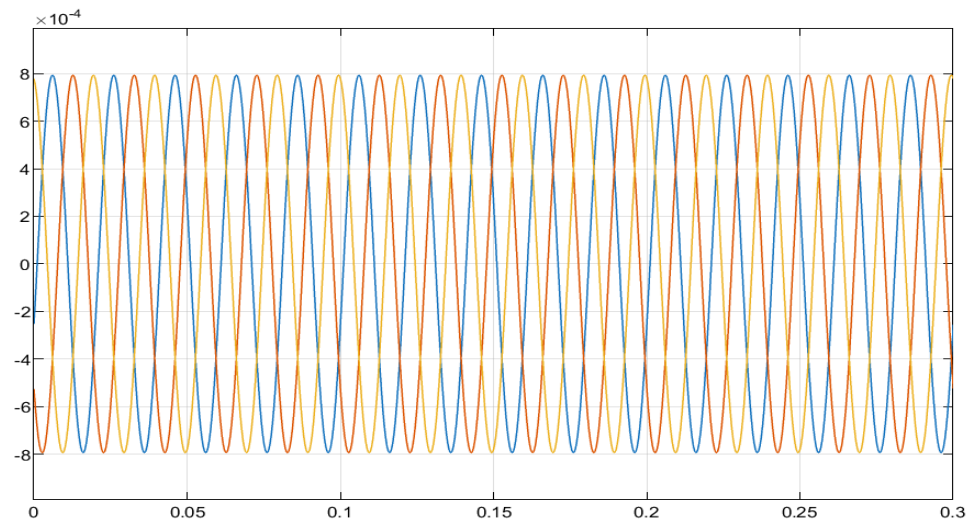


Figure 4. Secondary-Side Current I_2 during Normal Operation.

Figure 5. shows that the trip signal remains at 0 during normal operation. This means that the relay does not issue a trip command because no internal fault is detected.

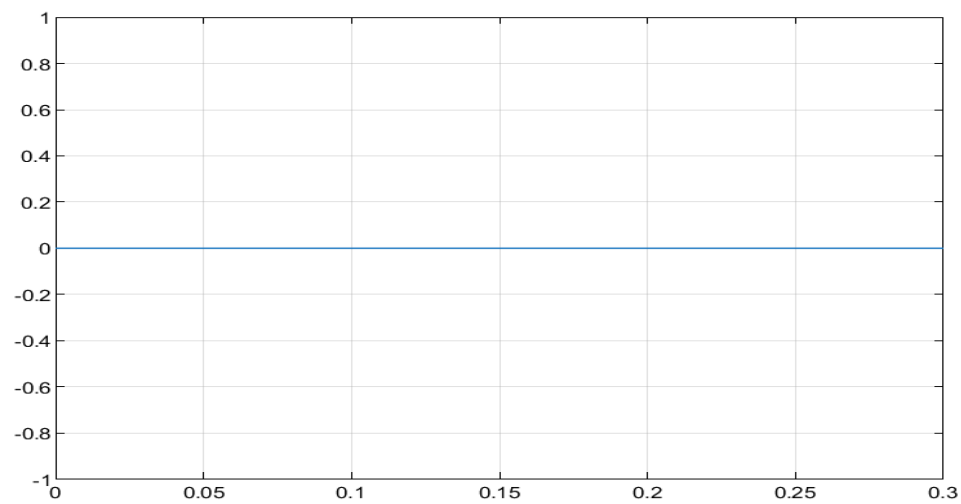


Figure 5. Trip Signal during Normal Operation.

Figure 6. shows that CB_{cmd} remains at 1 during normal operation. This means that the CB stays closed.

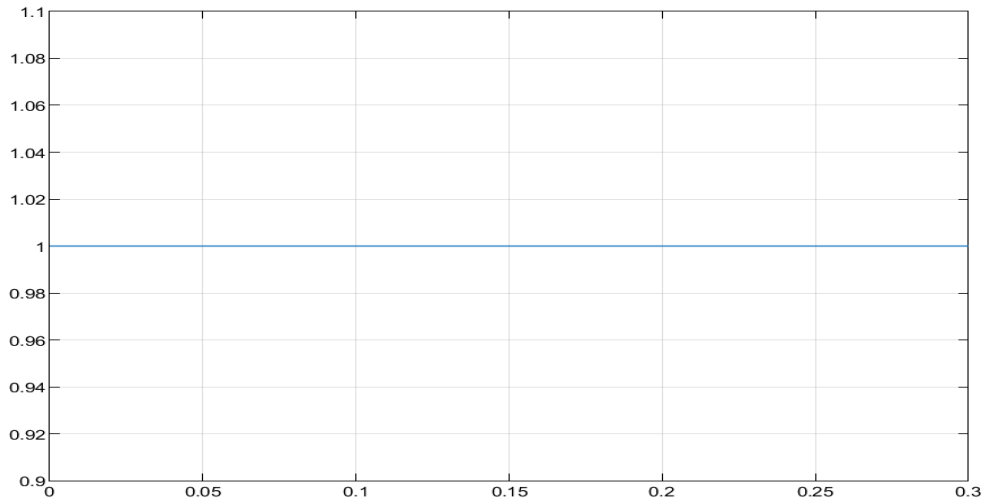


Figure 6. CB_{cmd} during Normal Operation.

6.2 Internal Fault

In this scenario an internal fault was applied inside the transformer protection zone at $t = 0.2\text{s}$. After the fault occurred the balance between the primary-side and secondary-side currents was disturbed causing the differential current to increase. As a result, the relay detected an internal fault trip changed from 0 to 1, CB_{cmd} changed from 1 to 0 which caused the CB to open and isolate the transformer.

Figure 7. shows the primary-side (I_1) current during the internal fault test. A clear transient appears at the fault instant around 0.2 s. After the relay operates and the CB opens the current decreases to zero confirming that the faulted transformer is isolated.

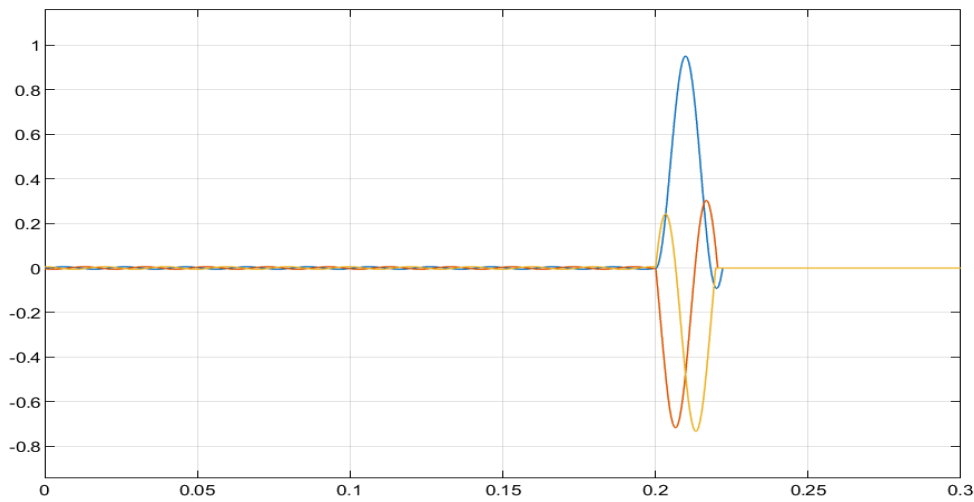


Figure 7. Primary-Side Current I_1 during Internal Fault.

Figure 8. presents the secondary-side (I_2) current during the internal fault test. Before the fault, the current waveform is stable and balanced. At about 0.2 s, the fault causes a disturbance in the current waveform. After the relay operates and the circuit breaker opens, the current drops to nearly zero, which confirms that the transformer is disconnected from the system.

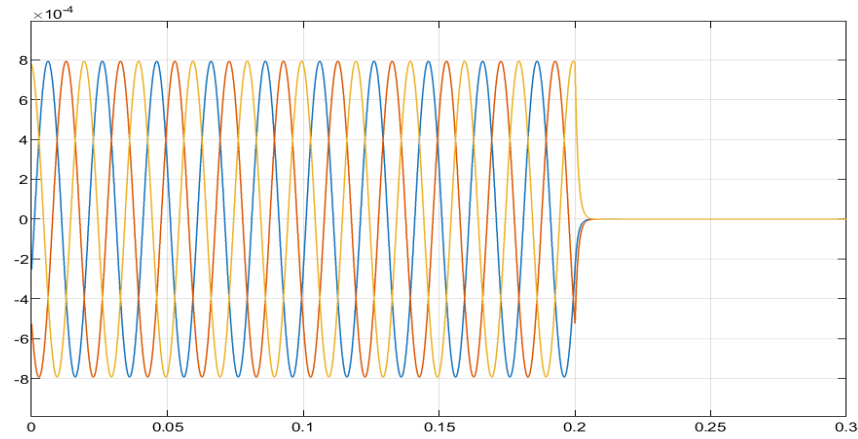


Figure 8. Secondary-Side Current I_2 during Internal Fault.

Figure 9. shows the differential current (I_d) response during the internal fault test. After the fault is applied at about 0.2s the differential current increases clearly above its normal value. This confirms that the relay detects an unbalance between the primary-side (I_1) and secondary-side (I_2) currents.

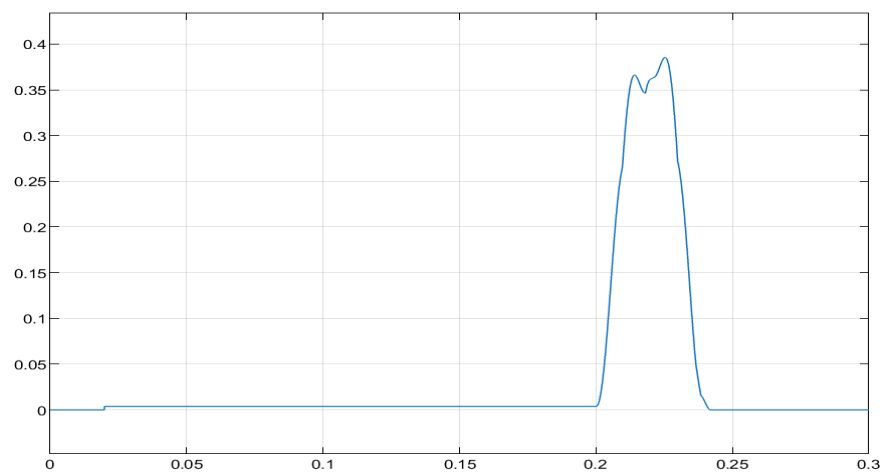


Figure 9. Differential Current I_d during Internal Fault.

Figure 10. shows that the trip signal changes from 0 to 1 after the internal fault is detected. This means that the relay correctly identifies the fault as an internal fault and issues a trip command.

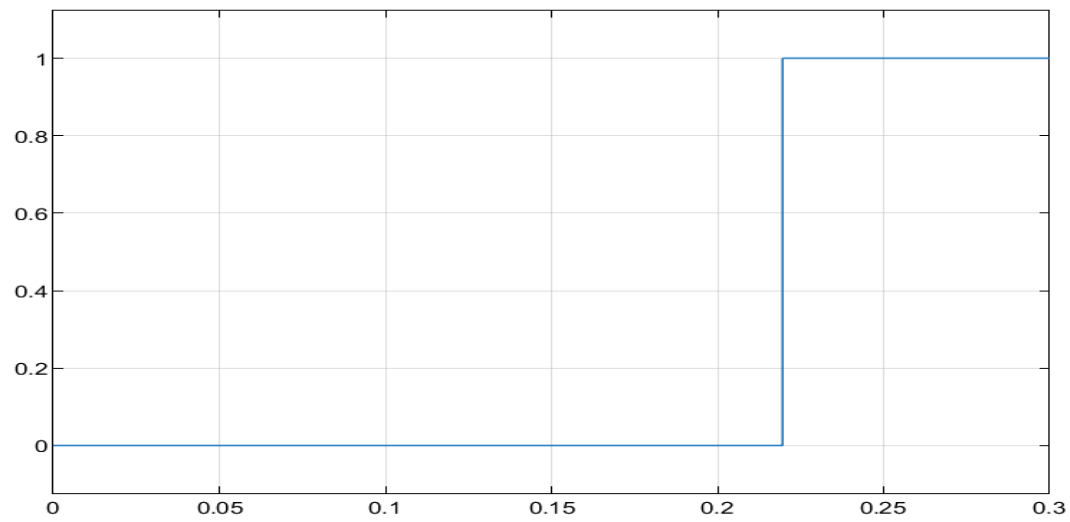


Figure 10. Trip Signal during Internal Fault.

Figure 11. Below shows that CB_{cmd} changes from 1 to 0 after the relay trip. Since the circuit breaker opens when $CB_{cmd} = 0$ this confirms that the transformer is isolated from the system after the internal fault.

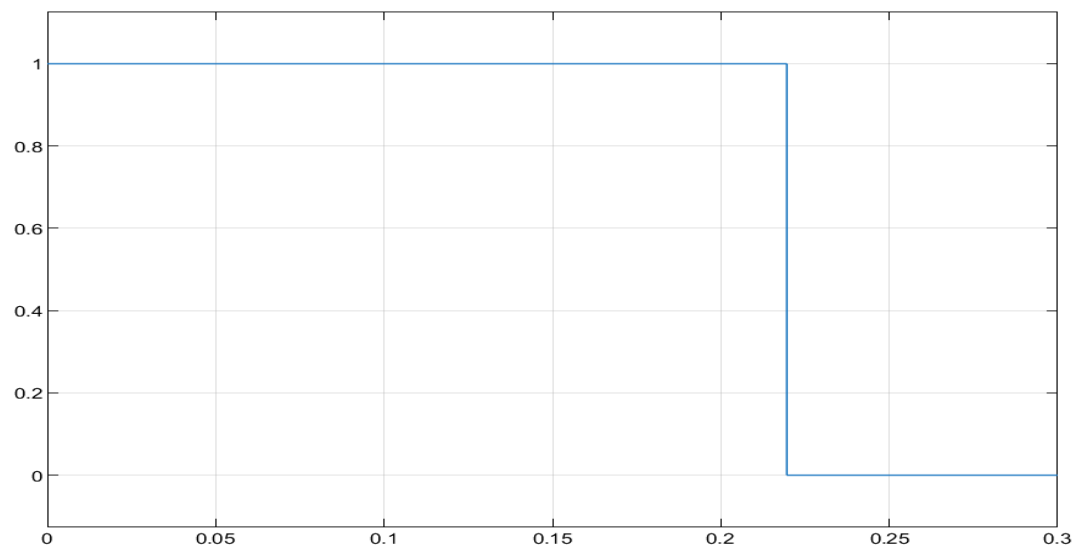


Figure 11. CB_{cmd} during Internal Fault.

6.3 External Fault

In this scenario the fault was applied outside the transformer protection zone. During the external fault, both the primary-side (I_1) and secondary-side (I_2) currents increased because the fault current passed through the transformer. The currents remained almost balanced from the relay point of view, so the differential current stayed small. As a result, trip remained 0 and CB_{cmd} remained 1 which means the circuit breaker stayed closed.

Figure 12. shows the primary-side current I_1 during the external fault test. At 0.2 s I_1 increases due to the external fault condition. The waveform becomes disturbed and its magnitude rises which indicates that the transformer is supplying fault current. Since the fault is outside the protected zone this condition does not cause relay tripping.

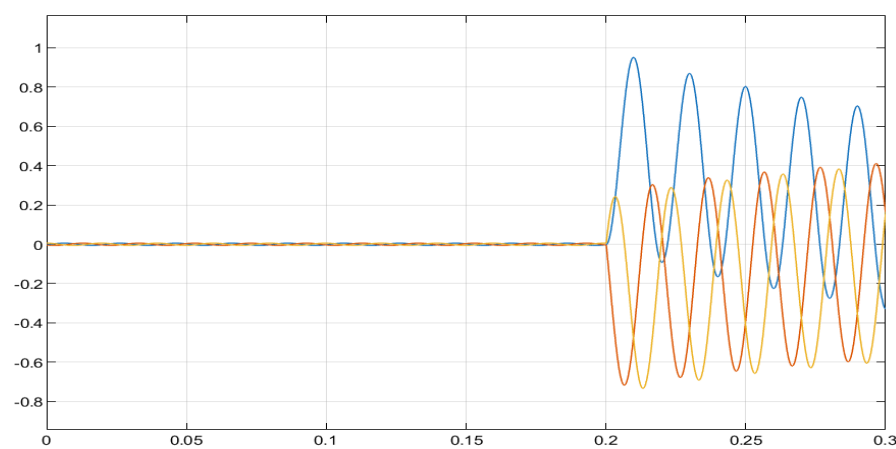


Figure 12. Primary-Side Current I_1 during External Fault

Figure 13. shows the secondary-side current I_2 during the external fault test. The waveform of I_2 also increases after the fault is applied like the primary-side current I_1 . This behavior confirms that the current is passing through the transformer toward the external fault rather than being caused by an internal transformer fault.

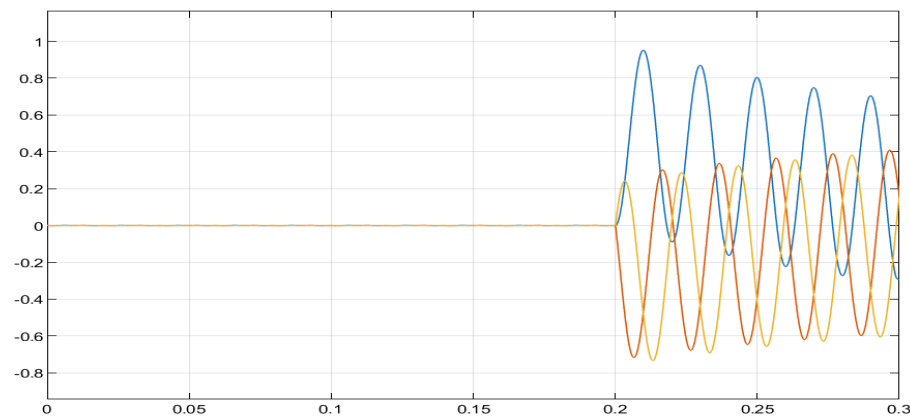


Figure 13. Secondary-Side Current I_2 during External Fault

Figure 14. presents that the differential current I_d remains very small during the external fault. Although the primary-side current I_1 and the secondary-side current I_2 increase the difference between them remains low. This confirms that the relay correctly identifies the fault as external and remains stable.

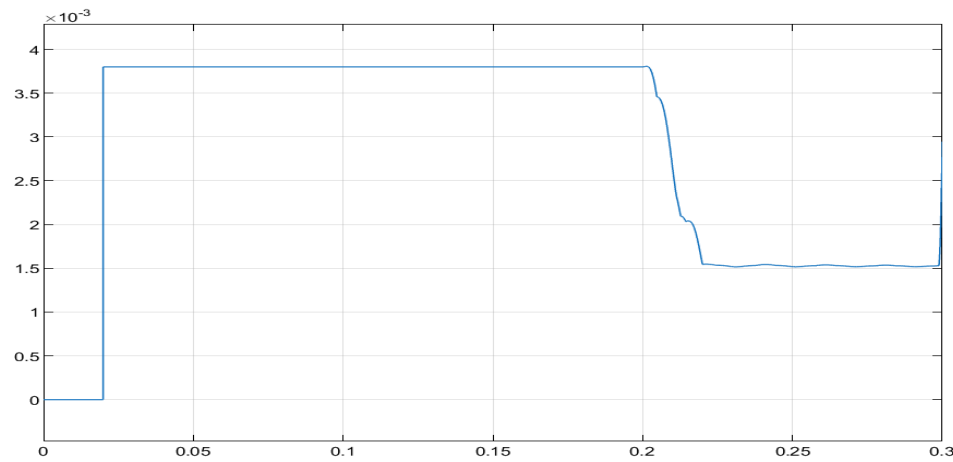


Figure 14. Differential Current I_d during External Fault.

Figure 15. shows that the Trip signal remains at 0 during the external fault. This means that the relay does not issue a trip command because the fault is outside the transformer protection zone.

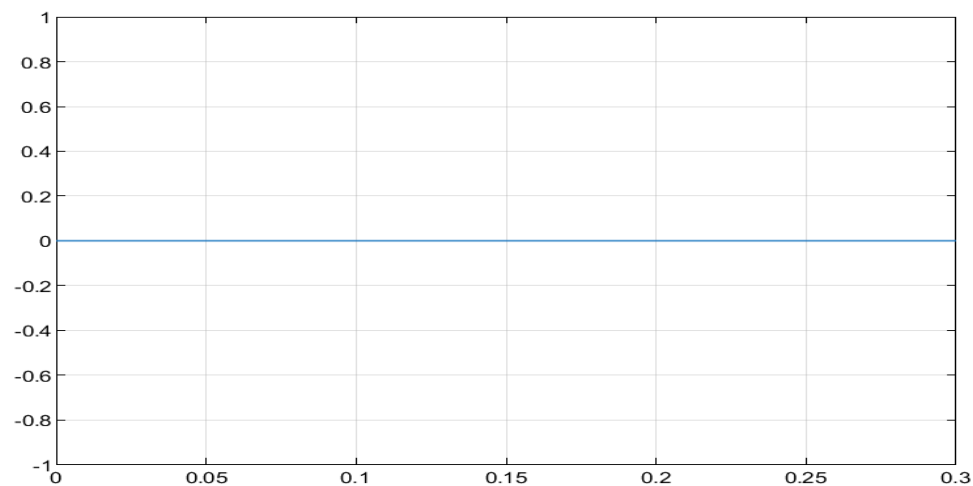


Figure 15. Trip Signal during External Fault.

Figure 16. shows that CB_{cmd} remains at 1 during the external fault. Since the CB stays closed when $CB_{cmd} = 1$ this confirms that the transformer is not disconnected for an external fault. This result shows that the relay operates selectively and avoids unnecessary breaker operation.

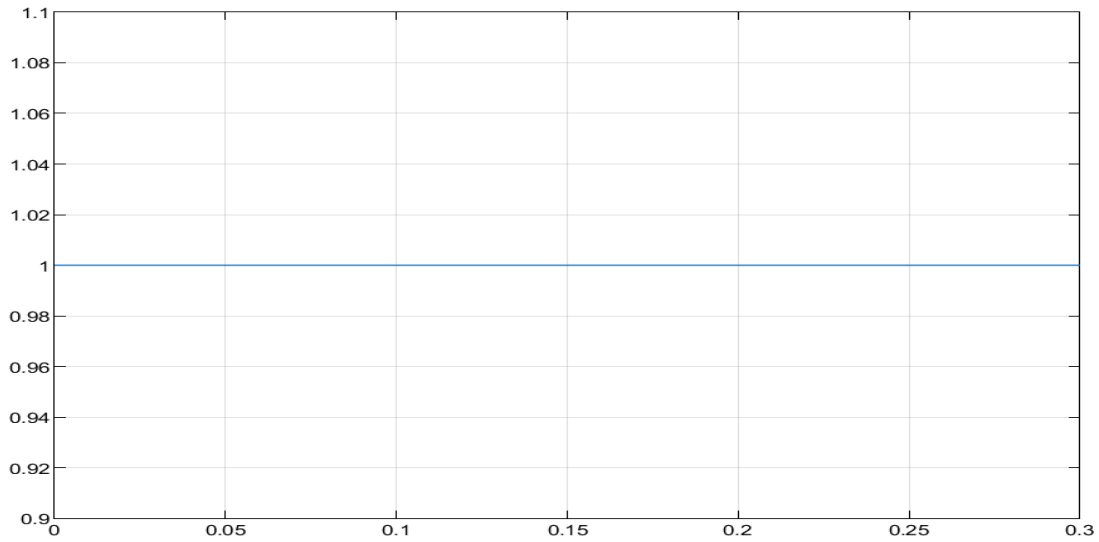


Figure 16. CB_{cmd} during External Fault.

6.4 Inrush Current Blocking

In this scenario, all fault blocks were disabled by setting the fault switching time to 10 which is outside the simulation time range. This ensures that no internal or external fault occurs during the test. To create the transformer energization condition, the source-side circuit breaker was initially open and then closed at approximately 0.1s. Therefore, the transformer was suddenly connected to the supply at this time.

Transformer saturation was enabled in the model to produce magnetizing inrush current during energization. The relay response in this test is related only to transformer energization, not to any fault condition. The purpose of this test is to verify that the relay can recognize the inrush condition using the second harmonic content and block false tripping, while keeping $Trip = 0$ and $CB_{cmd} = 1$.

Figure 17. shows the primary-side current during transformer energization with the fault blocks disabled. Before 0.1 s the current is zero because the transformer is disconnected from the source. At approximately 0.1s the source-side circuit breaker closes, and a magnetizing inrush current appears with a peaked and asymmetric waveform. This behavior is expected during transformer energization due to core saturation.

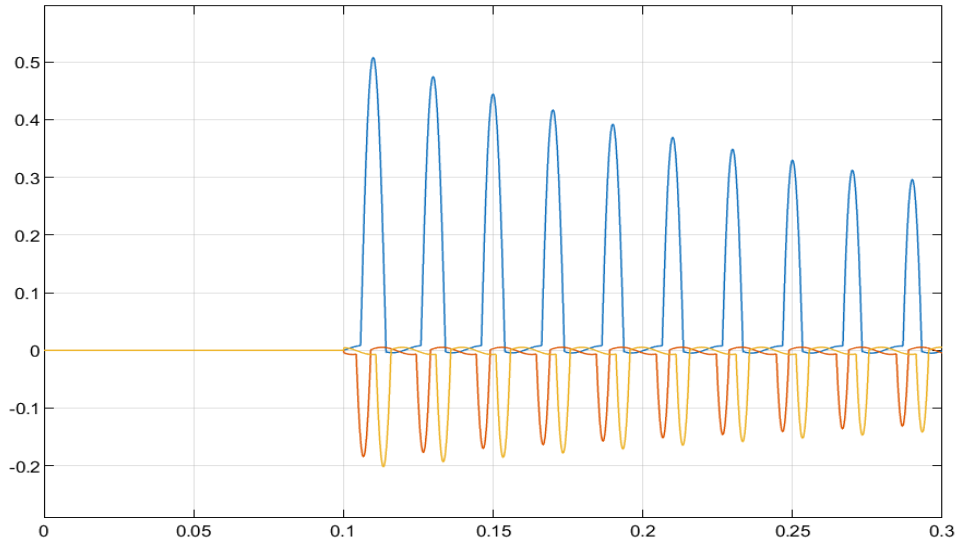


Figure 17. Primary-Side Current during Transformer Energization.

Figure 18. present the secondary-side current during transformer energization. Before 0.1 s the current was nearly zero because the transformer was disconnected from the source. At approximately 0.1s the source-side circuit breaker closes and the transformer is energized. After that, the secondary-side current appears and reaches a balanced sinusoidal waveform.

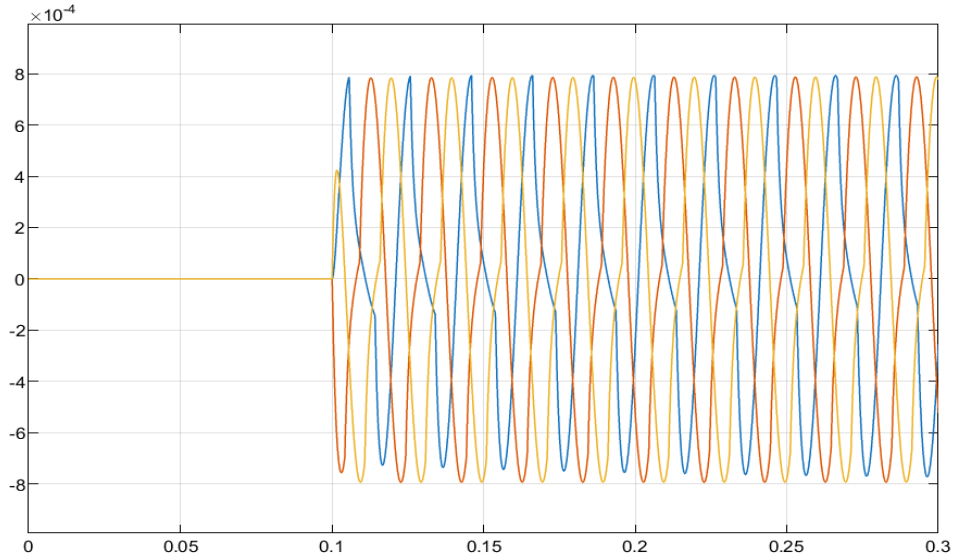


Figure 18. Secondary-Side Current during Transformer Energization.

Figure 19. presents the second harmonic ratio H_2 during transformer energization. At approximately 0.1s the transformer is energized and H_2 rises clearly. Since the second harmonic ratio is higher than the selected blocking threshold the relay identifies this condition as transformer inrush current. This prevents false tripping during transformer energization.

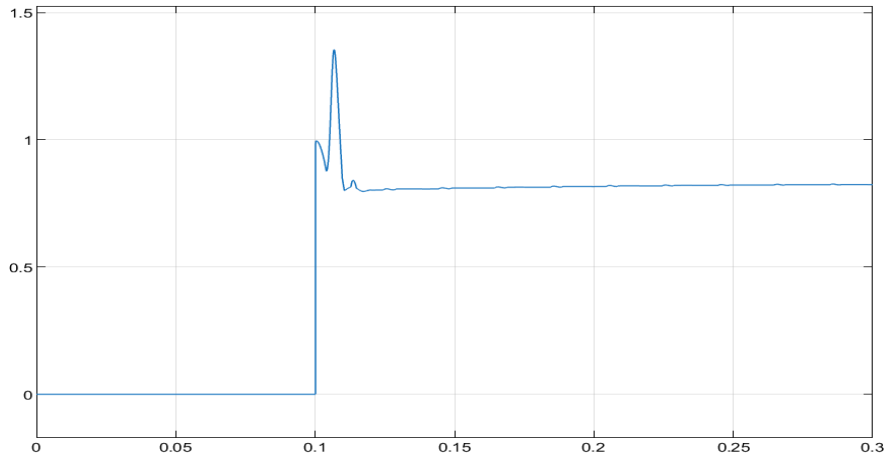


Figure 19. Second Harmonic Ratio H_2 during Transformer Energization.

Figure 20. shows the Inrush Block signal during transformer energization. At approximately 0.1s the signal changes from 0 to 1, which means that the relay detects the inrush condition and activates the blocking function. This prevents the relay from issuing a false trip during transformer energization.

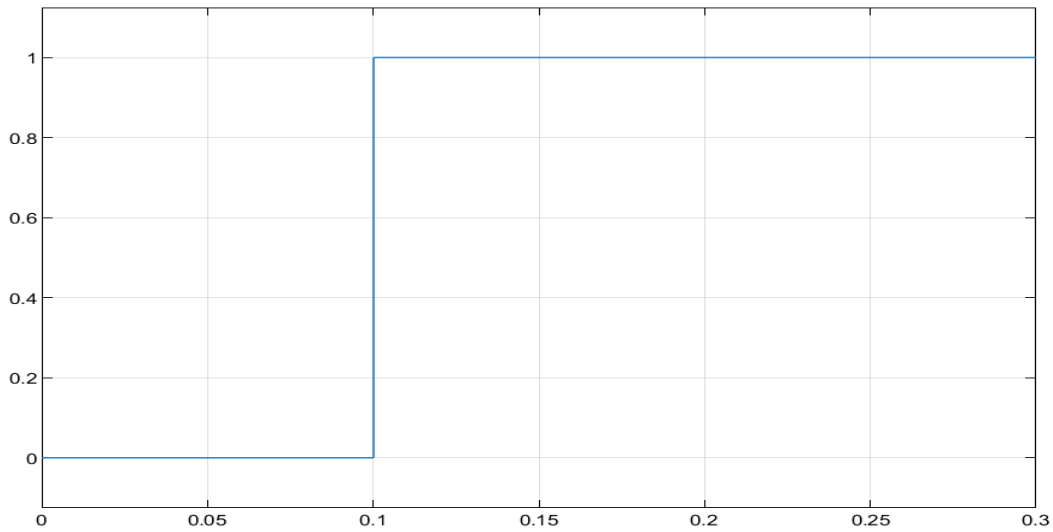


Figure 20. Inrush Blocking Signal during Transformer Energization.

Figure 21. presents the Trip signal during transformer energization. Although the second harmonic ratio increases and the inrush blocking signal becomes active the Trip signal remains at 0. This confirms that the relay does not operate during the inrush condition.

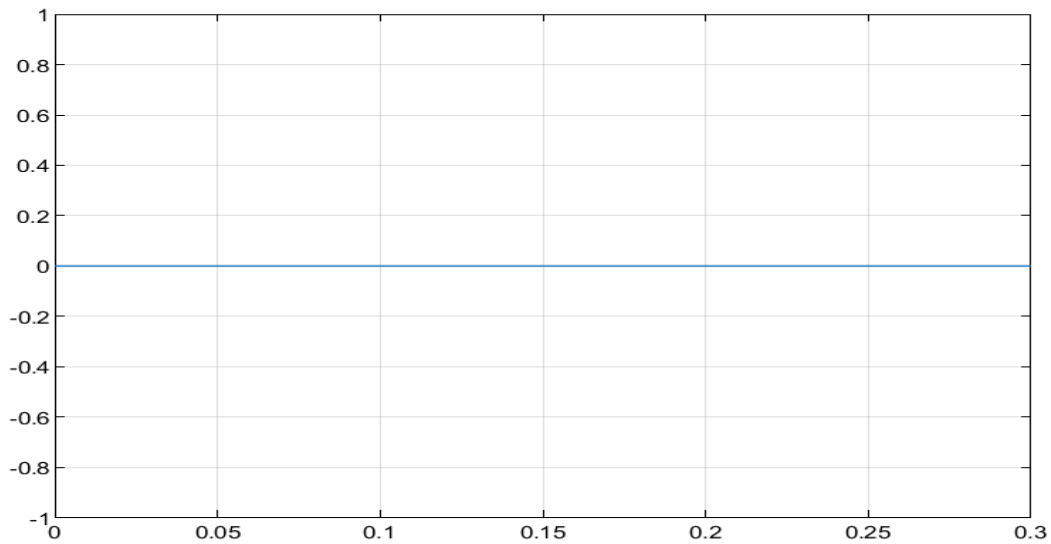


Figure 21. Trip Signal during Transformer Energization.

Figure 22. shows the circuit CB_{cmd} during transformer energization. The signal remains at 1 which means that the circuit breaker stays closed. This confirms that the relay remains stable and avoids unnecessary CB operation during transformer energization.

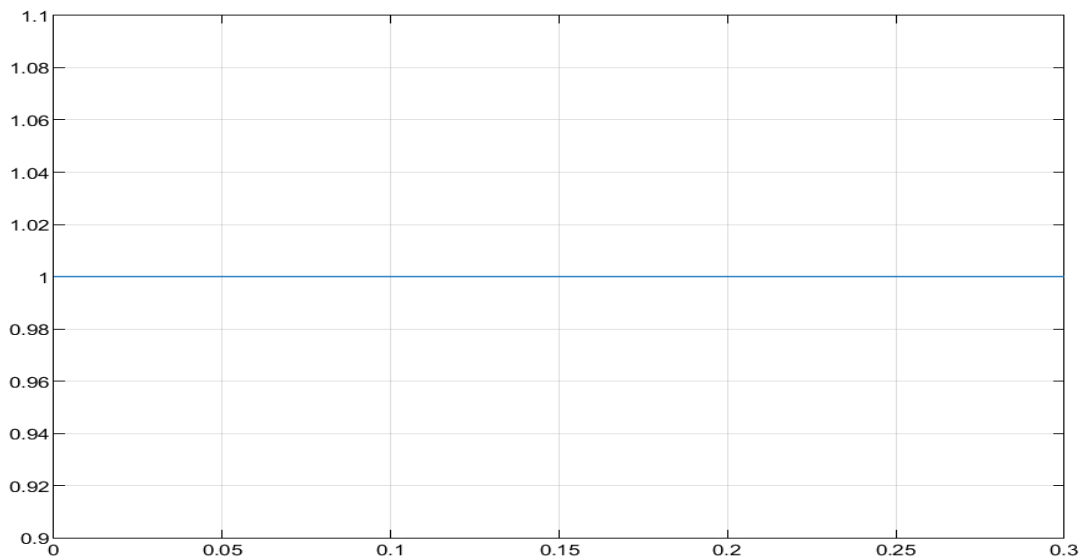


Figure 22. CB_{cmd} during Transformer Energization.

Figure 23. shows the inrush current indicator during transformer energization. Before 0.1 s, the current is nearly zero because the transformer is disconnected from the source. At approximately 0.1 s, the transformer is energized and a transient magnetizing inrush current appears. The current gradually decreases as the transformer approaches steady-state operation.

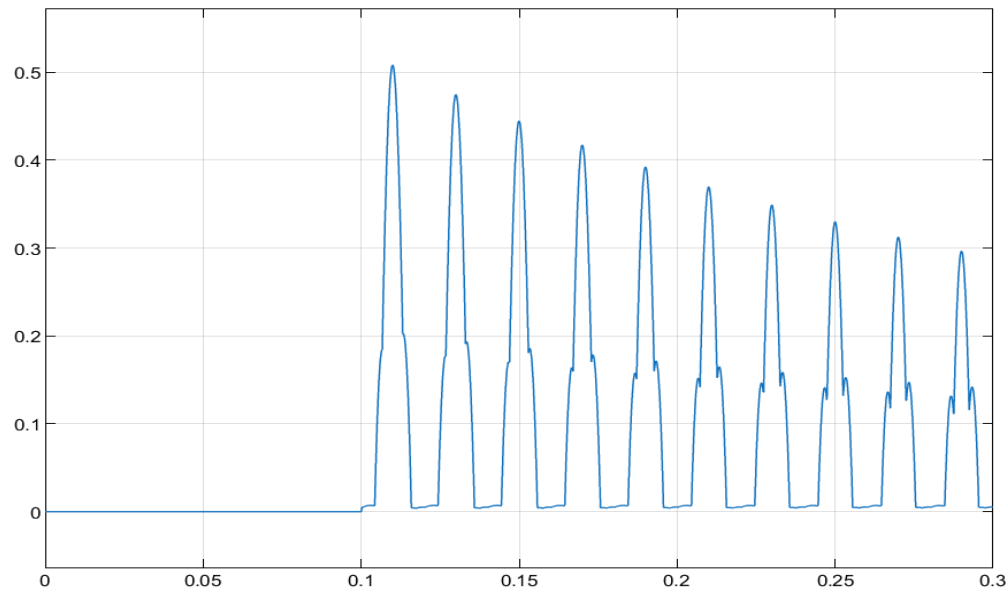


Figure 23. Inrush Current Indicator during Transformer Energization.

6.5 Single-Phase-to-Ground Internal Fault

In this test case, a single-phase-to-ground internal fault was applied inside the transformer protection zone. The fault was created by selecting Phase A and Ground only in the Three-Phase Fault block.

Figure 24. shows the primary-side current during the single-phase-to-ground internal fault. The waveform becomes disturbed mainly around the faulted phase after the fault is applied at 0.2s. This indicates that the system is affected by an unbalanced internal fault.

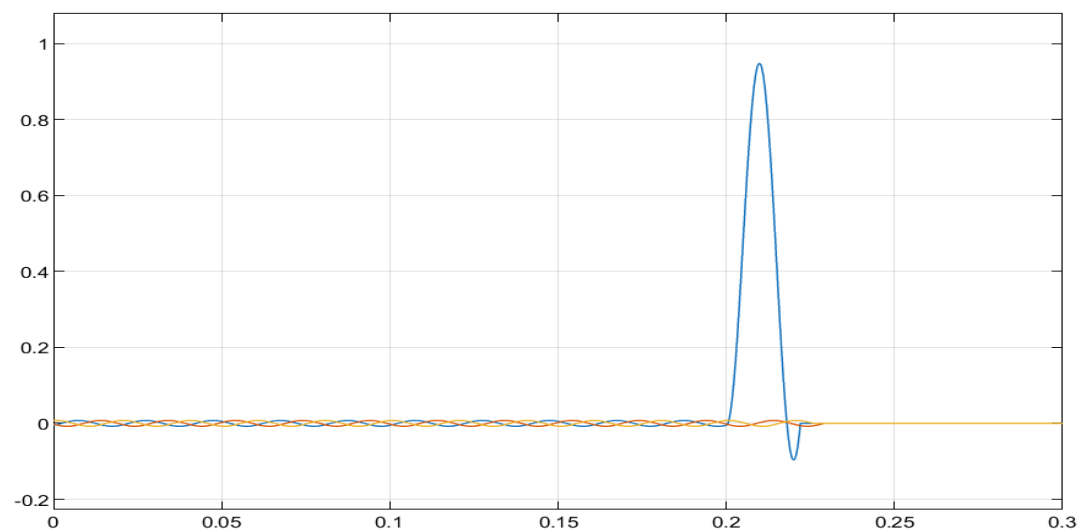


Figure 24. Primary-Side Current during Single-Phase-to-Ground Internal Fault.

Figure 25. shows the secondary-side current during the single-phase-to-ground internal fault. The current waveform becomes disturbed after the fault is applied. After the relay operates and the circuit breaker opens, the current decreases to nearly zero.

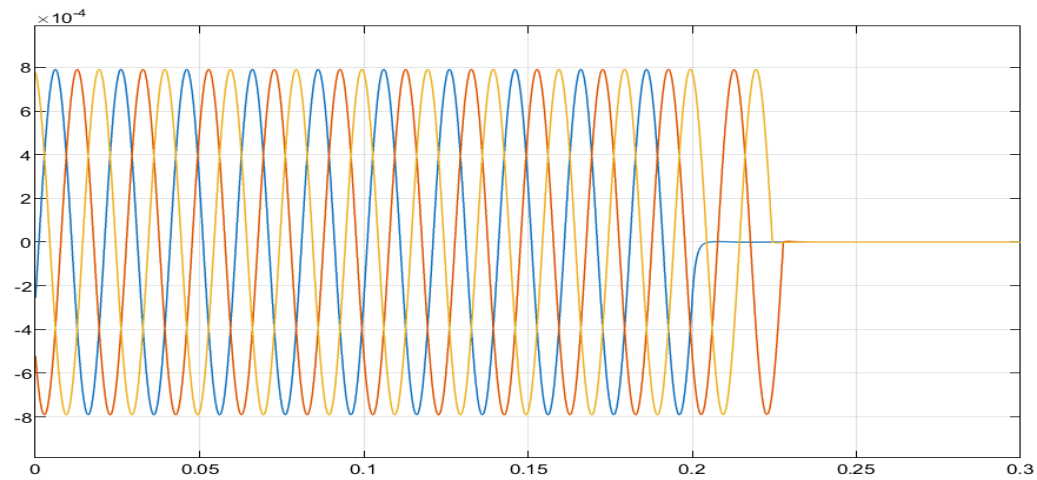


Figure 25. Secondary-Side Current during Single-Phase-to-Ground Internal Fault.

Figure 26. shows the trip signal response during the single-phase-to-ground internal fault test. Before the fault is applied, the trip signal remains at 0, indicating that the relay does not operate under normal conditions. After the fault occurs at approximately 0.2 s, the relay detects the internal fault, and the trip signal changes from 0 to 1. This confirms that the relay successfully identifies the single-phase-to-ground fault as an internal fault.

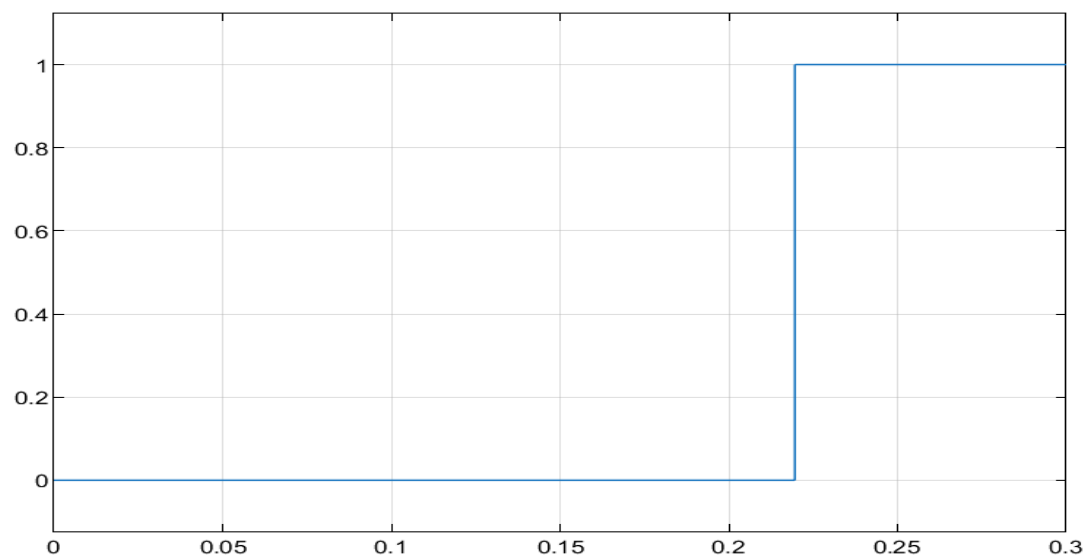


Figure 26. Trip Signal during Single-Phase-to-Ground Internal Fault.

Figure 27. shows the circuit breaker command during the single-phase-to-ground internal fault test. Before the fault occurs, CB_{cmd} remains at 1, indicating that the circuit breaker is closed. After the relay detects the internal fault, CB_{cmd} changes from 1 to 0, causing the circuit breaker to open and isolate the transformer from the system. This confirms the correct operation of the relay and breaker control logic during an unbalanced internal fault.

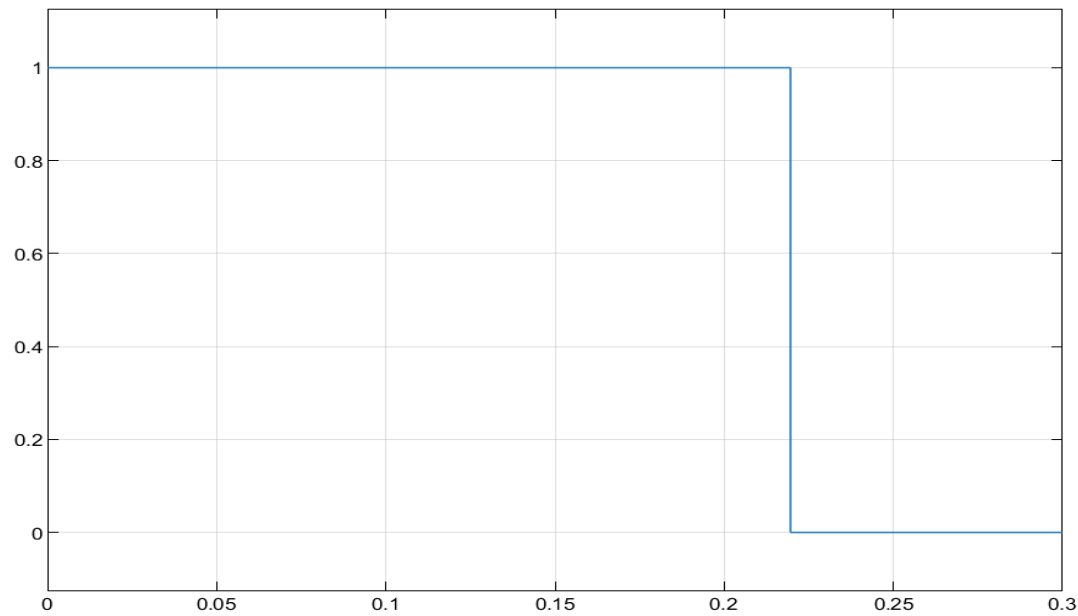


Figure 27. CB_{cmd} during Single-Phase-to-Ground Internal Fault.

Table A. summarizes the relay response under all simulated scenarios. The results show that the relay operates correctly for internal faults and remains stable during normal operation, external faults, and transformer energization. This confirms that the proposed differential relay can distinguish between fault and non-fault conditions.

Test Case	Fault Location / Condition	Expected Relay Response	Simulation Result
Normal Operation	No fault	No trip	Trip = 0, CB_{cmd} = 1
Three-Phase-to-Ground Internal Fault	Inside protection zone	Trip	Trip = 1, CB_{cmd} = 0
Three-Phase-to-Ground External Fault	Outside protection zone	No Trip	Trip = 0, CB_{cmd} = 1
Transformer Inrush Current	Transformer energization	No trip	increased, Inrush Block = 1, Trip = 0, CB_{cmd} = 1
Single-Phase-to-Ground Internal Fault	Inside protection zone	Trip	Trip = 1, CB_{cmd} = 0

Table A. Summary of Relay Response under Different Test Cases.

Conclusion

This project presented the design and simulation of a power transformer differential protection relay using MATLAB/Simulink. The relay was tested under different Scenarios including normal operation, internal faults, external faults, transformer energization and single-phase-to-ground faults.

The simulation results showed that the relay successfully detected internal faults and generated the appropriate trip signal to isolate the transformer. At the same time the relay remained stable during external faults and normal operating conditions. The second harmonic blocking method effectively prevented false tripping during transformer energization and magnetizing inrush current conditions.

The implemented relay algorithm demonstrated good selectivity, sensitivity and stability. The proposed differential relay model can provide reliable transformer protection under different power system Scenarios.

References

- [1] B. Kasztenny and M. Kezunovic, "Digital Relays Improve Protection of Large Power Transformer," *IEEE Computer Applications in Power*, vol. 11, no. 4, pp. 39–45, Oct. 1998, doi: 10.1109/67.721702.
Link: <https://doi.org/10.1109/67.721702>
- [2] B. Gao, W. Wei, L. Zhang, N. Chen, Y. Wu, and Y. Tang, "Differential Protection for an Outgoing Transformer of Large-Scale Doubly Fed Induction Generator-Based Wind Farms," *Energies*, vol. 7, no. 9, pp. 5566–5585, 2014, doi: 10.3390/en7095566.
Link: <https://www.mdpi.com/1996-1073/7/9/5566>
- [3] S. Key, G.-W. Son, and S.-R. Nam, "Deep Learning-Based Algorithm for Internal Fault Detection of Power Transformers during Inrush Current at Distribution Substations," *Energies*, vol. 17, no. 4, article 963, 2024, doi: 10.3390/en17040963.
Link: <https://www.mdpi.com/1996-1073/17/4/963>
- [4] R. Hamilton, "Analysis of Transformer Inrush Current and Comparison of Harmonic Restraint Methods in Transformer Protection," *IEEE Transactions on Industry Applications*, vol. 49, no. 4, pp. 1890–1899, 2013, doi: 10.1109/TIA.2013.2257155.
Link: <https://doi.org/10.1109/TIA.2013.2257155>
- [5] A. Azeem, M. Jamil, S. Qamar, H. Malik, and R. A. Thokar, "Design of Hardware Setup Based on IEC 61850 Communication Protocol for Detection & Blocking of Harmonics in Power Transformer," *Energies*, vol. 14, no. 24, article 8284, 2021, doi: 10.3390/en14248284.
Link: <https://doi.org/10.3390/en14248284>
- [6] MathWorks, Inc., *MATLAB and Simulink R2024b*, Natick, MA, USA, 2024.
[Online]. Available: <https://www.mathworks.com/products/matlab.html>

Appendix. Additional Simulation Results

Figure A1. shows the restraint current (I_r) during the internal fault scenario. The restraint current increases as the through-fault current increases which helps the relay remain stable and avoid unwanted tripping.

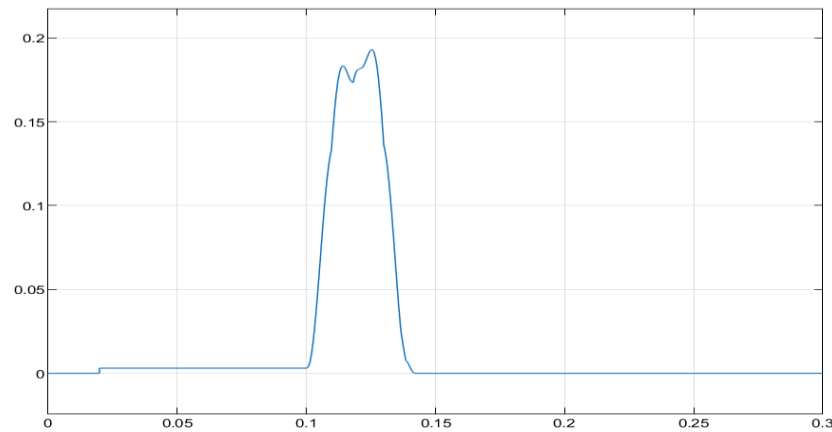


Figure A1. Restraint Current I_r during Internal Fault.

Figure A2. shows the primary-side and secondary-side voltage waveform during the internal fault scenario. The voltage remains sinusoidal before the fault then becomes disturbed when the internal fault occurs. After the relay trips and the circuit breaker opens the voltage drops to nearly zero.

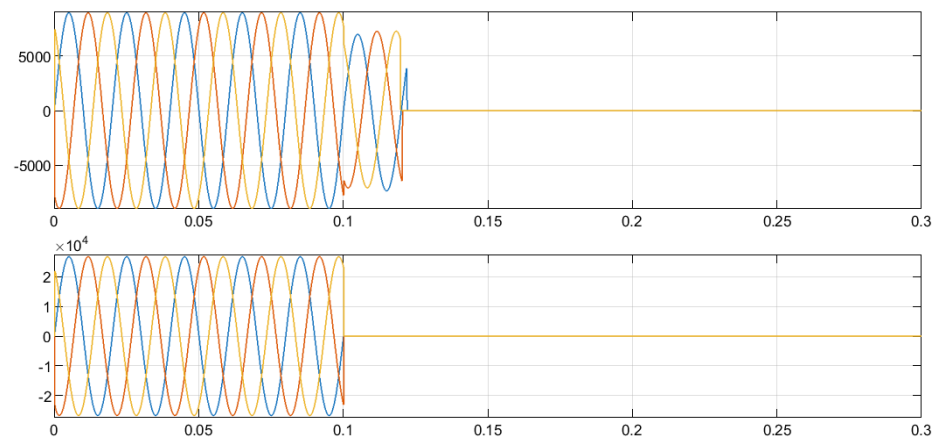


Figure A2. Primary-Side and secondary-side Voltage during Internal Fault.

